

SMART LENS FOR VISION ASSISTANCE

PROJECT REPORT

ABSTRACT

The Smart Lens for Vision Assistance project is a software-based assistive system designed to help users, especially visually impaired individuals, read printed text. The system captures real-time video through a webcam, processes the image to enhance text clarity, extracts text using Optical Character Recognition (OCR), and converts the detected text into speech. This project demonstrates the practical application of computer vision and speech synthesis using Python.

1.1 Overview

Advancements in computer vision and artificial intelligence have enabled the development of assistive technologies for people with visual impairments. The Smart Lens for Vision Assistance project aims to provide real-time text reading support using a camera and software-based processing.

1.2 Problem Statement

Visually impaired individuals often face difficulty reading printed text such as books, labels, or notices. Existing solutions are either expensive or hardware-dependent. A low-cost software solution can improve accessibility.

1.3 Objectives

- To capture live video using a webcam
 - To detect and extract text from images
 - To convert detected text into speech
 - To assist visually impaired users
-

Previous assistive systems relied heavily on dedicated hardware such as smart glasses. Recent research shows that computer vision combined with OCR and text-to-speech can provide effective vision assistance using standard cameras and software tools.

3.1 Hardware Requirements

- Laptop or Desktop Computer
- Built-in or external webcam

3.2 Software Requirements

- Windows Operating System
 - Python 3.13 or above
 - OpenCV Library
 - Tesseract OCR Engine
 - PyTesseract
 - PyTTsX3
-

4.1 Block Diagram Description

1. Camera Input
2. Image Preprocessing
3. OCR Text Detection
4. Text-to-Speech Conversion
5. Audio Output

4.2 Working Principle

The camera captures live frames which are converted to grayscale and processed using thresholding techniques. OCR is applied to extract text, which is then converted into speech using a text-to-speech engine.

Step 1: Camera Capture

OpenCV is used to access the webcam and capture real-time frames.

Step 2: Image Preprocessing

The captured image is converted to grayscale, blurred, and thresholded to enhance text visibility.

Step 3: Text Detection

Tesseract OCR is used to extract text from the processed image.

Step 4: Text-to-Speech

The detected text is converted into audible speech using PyTTsX3.

6.1 Programming Language

Python

6.2 Libraries Used

- OpenCV
- PyTesseract
- PyTTSx3

Source Code – Step by Step

Step 1: Import Required Libraries

```
import cv2
import pytesseract
import pyttsx3
```

Step 2: Configure Tesseract OCR Path

```
pytesseract.pytesseract.tesseract_cmd = r"C:\Program Files\Tesseract-OCR\tesseract.exe"
```

Step 3: Initialize Text-to-Speech Engine

```
engine = pyttsx3.init()
engine.setProperty('rate', 150)
```

Step 4: Initialize Webcam

```
cap = cv2.VideoCapture(0)
if not cap.isOpened():
    print("Cannot open camera")
    exit()
```

Step 5: Capture Frames and Preprocess Image

```
ret, frame = cap.read()
frame = cv2.resize(frame, (640, 480))
gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
blur = cv2.GaussianBlur(gray, (5, 5), 0)
thresh = cv2.adaptiveThreshold(
    blur, 255,
    cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
    cv2.THRESH_BINARY, 11, 2
)
```

Step 6: Perform OCR and Convert Text to Speech

```
text = pytesseract.image_to_string(thresh).strip()
if text != "":
    print("Detected Text:", text)
    engine.say(text)
    engine.runAndWait()
```

Step 7: Display Output and Exit Control

```
cv2.imshow("Smart Lens OCR", frame)
cv2.imshow("Thresholded", thresh)
if cv2.waitKey(1) & 0xFF == ord('q'):
    cap.release()
    cv2.destroyAllWindows()
```

Final Source Code

```
python
import cv2
import pytesseract
import pyttsx3

pytesseract.pytesseract.tesseract_cmd = r"C:\\Program Files\\Tesseract-OCR\\
\\tesseract.exe"

engine = pyttsx3.init()
engine.setProperty('rate', 150)

cap = cv2.VideoCapture(0)

if not cap.isOpened():
    print("Cannot open camera")
    exit()

last_text = ""

while True:
    ret, frame = cap.read()
    if not ret:
        break

    frame = cv2.resize(frame, (640, 480))
```

```
gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
blur = cv2.GaussianBlur(gray, (5, 5), 0)
thresh = cv2.adaptiveThreshold(blur, 255,
                               cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
                               cv2.THRESH_BINARY, 11, 2)

text = pytesseract.image_to_string(thresh).strip()

if text != "" and text != last_text:
    print("Detected Text:", text)
    engine.say(text)
    engine.runAndWait()
    last_text = text

cv2.imshow("Smart Lens OCR", frame)
cv2.imshow("Thresholded", thresh)

if cv2.waitKey(1) & 0xFF == ord('q'):
    break

cap.release()
cv2.destroyAllWindows()
```

The system successfully detects printed text from live camera input and converts it into audible speech. The output is clear and works in real time.

- Assistive technology for visually impaired users
 - Smart reading devices
 - Educational tools
-

The Smart Lens for Vision Assistance project demonstrates the effective use of computer vision and OCR for real-time text recognition and speech conversion. The system is cost-effective, easy to use, and suitable for academic and assistive applications.

- Multi-language text recognition
 - Object detection integration
 - Mobile or wearable implementation
 - Voice command support
-

REFERENCES

1. OpenCV Documentation
2. Tesseract OCR Documentation
3. Python Official Documentation